SECTION 2

LIMITATIONS

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SECTION 2

LIMITATIONS

WARNING WHEN ANY LIMITATION IS EXCEEDED, MAINTENANCE ACTION MAY BE REQUIRED AND NECESSARY BEFORE NEXT FLIGHT. ENTER DURATION AND AMOUNT OF EXCESS IN LOGBOOK.

2.1 GENERAL

This helicopter shall be operated in compliance with the limitations of this section.

The "Airworthiness Limitations" in Chapter 101 of the BO105 Rotorcraft Maintenance Manual must be complied with.

For definitions of terms, abbreviations and symbols used in this section refer to Section 1.

2.2 KINDS OF OPERATION

The helicopter in its basic configuration is approved for operation under day and night Visual Flight Rules.

With special equipment installed and operative, and under observance of the procedures and limitations described in FMS 11-2, the helicopter is also certified for operation according to Instrument Flight Rules.

2.3 BASIS OF CERTIFICATION

This helicopter is certified by the LBA in the Normal Category based on FAR PART 27, Amendments 27-1 through 27-3.

2.4 OCCUPANTS

EFFECTIVITY *Model C Variants*

The helicopter in its basic configuration is approved as a five-place rotorcraft.

EFFECTIVITY *Model S Variants*

The helicopter in its basic configuration is approved as a five-place rotorcraft with an optional six-place configuration.

EFFECTIVITY *ALL*

2.4.1 Flight Crew

The minimum flight crew consists of one pilot in the right crew seat.

2.5 FLIGHT WITH OPTIONAL EQUIPMENT INSTALLED

Refer to Section 10 for additional limitations, procedures and performance data.

2.6 MASS AND LOAD LIMITATIONS

2.6.1 Maximum Gross Mass

Maximum approved gross mass .... 2400 kg

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*

Maximum approved gross mass .... 2500 kg

EFFECTIVITY *ALL*

2.6.2 Minimum Gross Mass

Minimum approved mass .... 1140 kg

2.6.3 Loading

Maximum cargo load is limited by Mass and Balance consideration.

Maximum allowable floor loading .... 600 kg/m^2

Max allowable load avionics equipment bay .... 20 kg

CAUTION CARGO, BAGGAGE AND LOOSE ITEMS MUST BE PROPERLY STOWED AND TIED DOWN IN ORDER TO MAKE IN-FLIGHT SHIFTING IMPOSSIBLE (REFER TO SECTION 6).

2.6.4 Cargo Securing Points

Maximum allowable load per:

- Tie-down ring .... 100 kg
- Insert .... 200 kg

2.7 CENTER OF GRAVITY LIMITATIONS

2.7.1 Longitudinal Center of Gravity

Station zero (datum) is an imaginary vertical plane, perpendicular to the aircraft centerline and located 3000 mm forward of the leveling point (see Section 6).

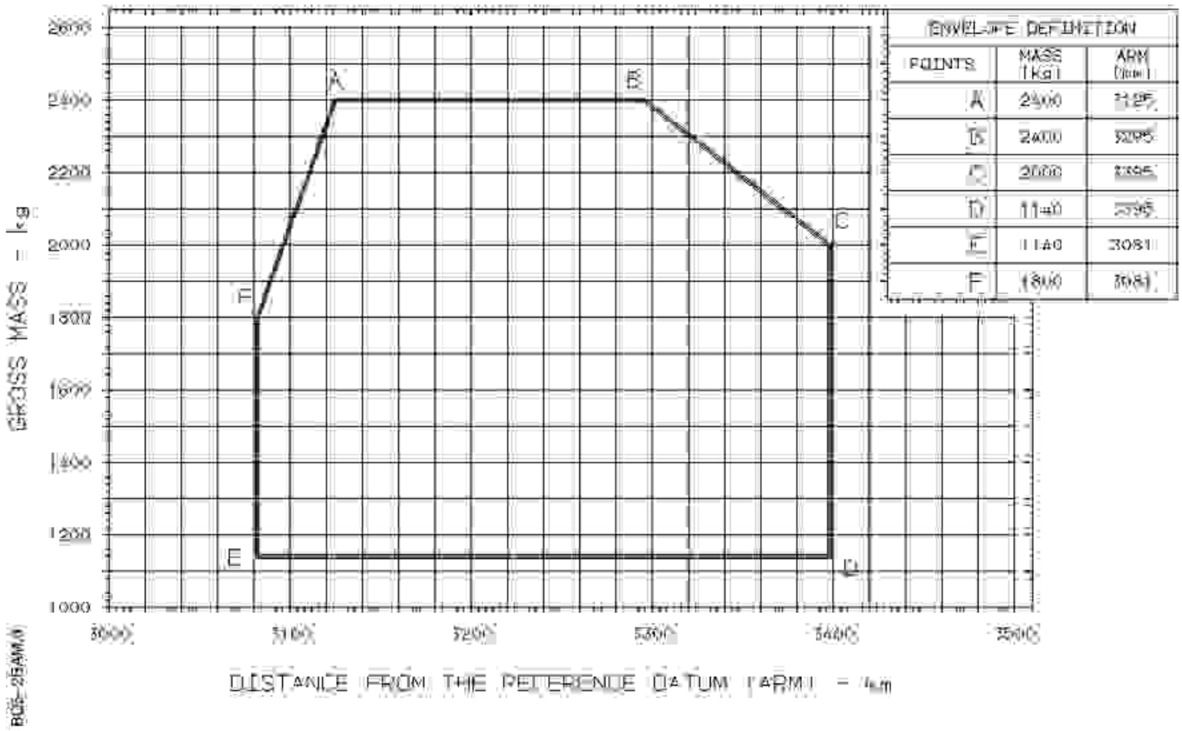


Fig. 2-1 Allowable CG Envelope (Longitudinal)

2.7.2 Lateral Center of Gravity

Lateral center of gravity limits left and right of the fuselage centerline . . . . 100 mm

2.8 AIRSPEED LIMITATIONS

2.8.1 Forward Flight

The following (Fig. 2-2) shows the airspeed limits for any gross mass up to 2400 kg.

PRESSURE ALTITUDE-FT	OAT °C									
	-45	-30	-20	-10	0	+10	+20	+30	+40	+50
0	145	145	145	145	145	145	144	141	139	137
2000	145	145	145	145	144	141	139	136	134	132/123
4000	145	145	143	141	139	136	134/132	132/128	125/123	
6000	145	141	138	136	134/132	131/127	129/122	127/118		
8000	140	136	133/132	131/127	129/122	126/117	123/113	119/108		
10000	135/134	131/127	128/122	126/117	122/112	118/107	114/103			
12000	130/124	126/117	123/118	117/107	113/102	108/97	106/93			
14000	124/113	117/106	112/102	108/97	103/92	99/88	NOTE MAINTAIN IN SPEED RANGE FROM VNE MINUS 20KIAS TO VNE 98÷102% ROTOR RPM			
16000	114/103	107/98	103/91	98/87	93/82	89/78				
17000	109/98	102/91	98/86	93/81	88/77					
	VNE-KIAS GROSS WEIGHT (kg) 2100/2400									

B05-2B2M_0

SINGLE-VALUE BOXES: V_{NE} VALUE FOR GROSS MASS UP TO 2400 KG

DOUBLE-VALUE BOXES: LEFT VALUE FOR GROSS MASS UP TO 2100 KG
RIGHT VALUE FOR GROSS MASS ABOVE 2100 KG

Fig. 2-2 Never-exceed Speed (V_{NE})

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

The never-exceed speed (V_{NE}) is 10% less than V_{NE} 2400kg at any pressure altitude and temperature.

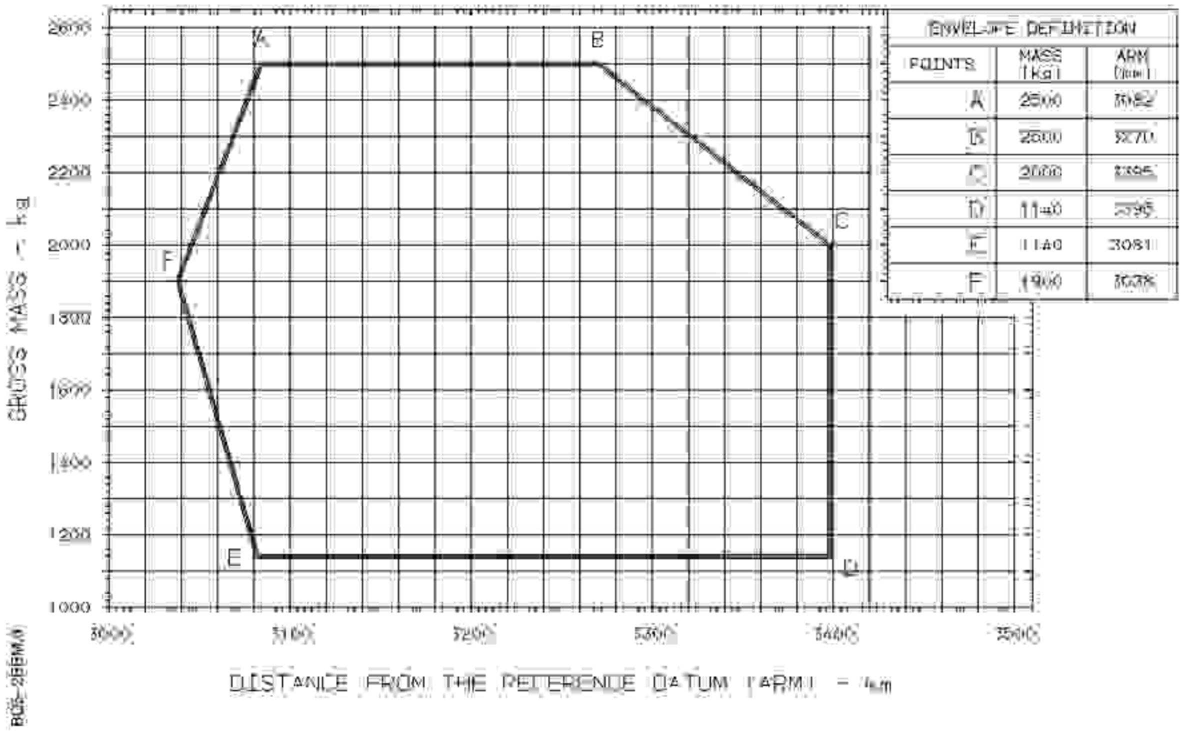
EFFECTIVITY *ALL*

The max airspeed for steady autorotation is 100 KIAS or that in accordance with the V_{NE} table (see Fig. 2-2) whichever value is less.

2.7 CENTER OF GRAVITY LIMITATIONS

2.7.1 Longitudinal Center of Gravity

Station zero (datum) is an imaginary vertical plane, perpendicular to the aircraft centerline and located 3000 mm forward of the leveling point (see Section 6).



ENVELOPE DEFINITION		
POINTS	MASS [Kg]	ARM [mm]
A	2500	3082
B	2500	3270
C	2000	3395
D	1140	3395
E	1140	3081
F	1900	3038

Fig. 2-1 Allowable CG Envelope (Longitudinal)

2.7.2 Lateral Center of Gravity

Lateral center of gravity limits left and right of the fuselage centerline . . . . 100 mm

EFFECTIVITY *When GM is between 2400 and 2500 kg*

Lateral center of gravity limits left and right of the fuselage centerline . . . . 80 mm

EFFECTIVITY *ALL*

2.8 AIRSPEED LIMITATIONS

2.8.1 Forward Flight

The following (Fig. 2-2) shows the airspeed limits for any gross mass up to 2400 kg.

PRESSURE ALTITUDE FT	OAT °C -45	-30	-20	-10	0	+10	+20	+30	+40	+50
0	145	145	145	145	145	145	144	141	139	137
2000	145	145	145	145	144	141	139	136	134	132/123
4000	145	145	143	141	139	136	134/132	132/128	125/123	
6000	145	141	138	136	134/132	131/127	129/122	127/113		
8000	140	136	133/132	131/127	129/122	126/117	123/113	119/108		
10000	135/134	131/127	128/122	126/117	122/112	118/107	114/103			
12000	130/124	126/117	123/118	117/107	113/102	108/97	106/93			
14000	124/113	117/106	112/102	108/97	103/92	99/88	NOTE MAINTAIN IN SPEED RANGE FROM VNE MINUS 20KTS TO VNE 98-102% ROTOR RPM			
16000	114/103	107/98	103/91	98/87	93/82	89/78				
17000	109/98	102/91	98/86	93/81	88/77					
	VNE-KIAS GROSS WEIGHT (kg) 2100/2400									

BO5-2&2M_0

SINGLE-VALUE BOXES: V_{NE} VALUE FOR GROSS MASS UP TO 2400 KG

DOUBLE-VALUE BOXES: LEFT VALUE FOR GROSS MASS UP TO 2100 KG
RIGHT VALUE FOR GROSS MASS ABOVE 2100 KG

Fig. 2-2 Never-exceed Speed (V_{NE})

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

The never-exceed speed (V_{NE}) is 10% less than V_{NE} 2400kg at any pressure altitude and temperature.

EFFECTIVITY *ALL*

The max airspeed for steady autorotation is 100 KIAS or that in accordance with the V_{NE} table (see Fig. 2-2) whichever value is less.

2.8.2 **Sideward Flight**

Sideward flight or crosswind hover has been demonstrated up to 25 kt airspeed.

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

Sideward flight or crosswind hover has been demonstrated up to 17 kt airspeed.

Engine anti-icing (and bleed air heating, if installed) must be switched off.

EFFECTIVITY *ALL*

2.8.3 **Rearward Flight**

Rearward flight or tailwind hover has been demonstrated up to 20 kt airspeed.

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

Rearward flight or tailwind hover has been demonstrated up to 17 kt airspeed.

Engine anti-icing (and bleed air heating, if installed) must be switched off.

EFFECTIVITY *ALL*

2.9 **ALTITUDE LIMITATIONS** (see Fig. 2-3)

NOTE Altitude values used throughout this manual are Pressure Altitude unless stated otherwise.

Maximum altitude:

– when operating on MIL-T-5624, grade JP-5 or ASTM D-1655 Jet A and Jet A1 .... 17,000 ft

– when operating on MIL-T-5624, grade JP-4; MIL-T-83133, grade JP-8; ASTM D-1655 Jet B; or alternate fuel (AVGAS/Jet fuel mixture) .... 13,700 ft

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

Maximum altitude .... 10,000 ft

EFFECTIVITY *ALL*

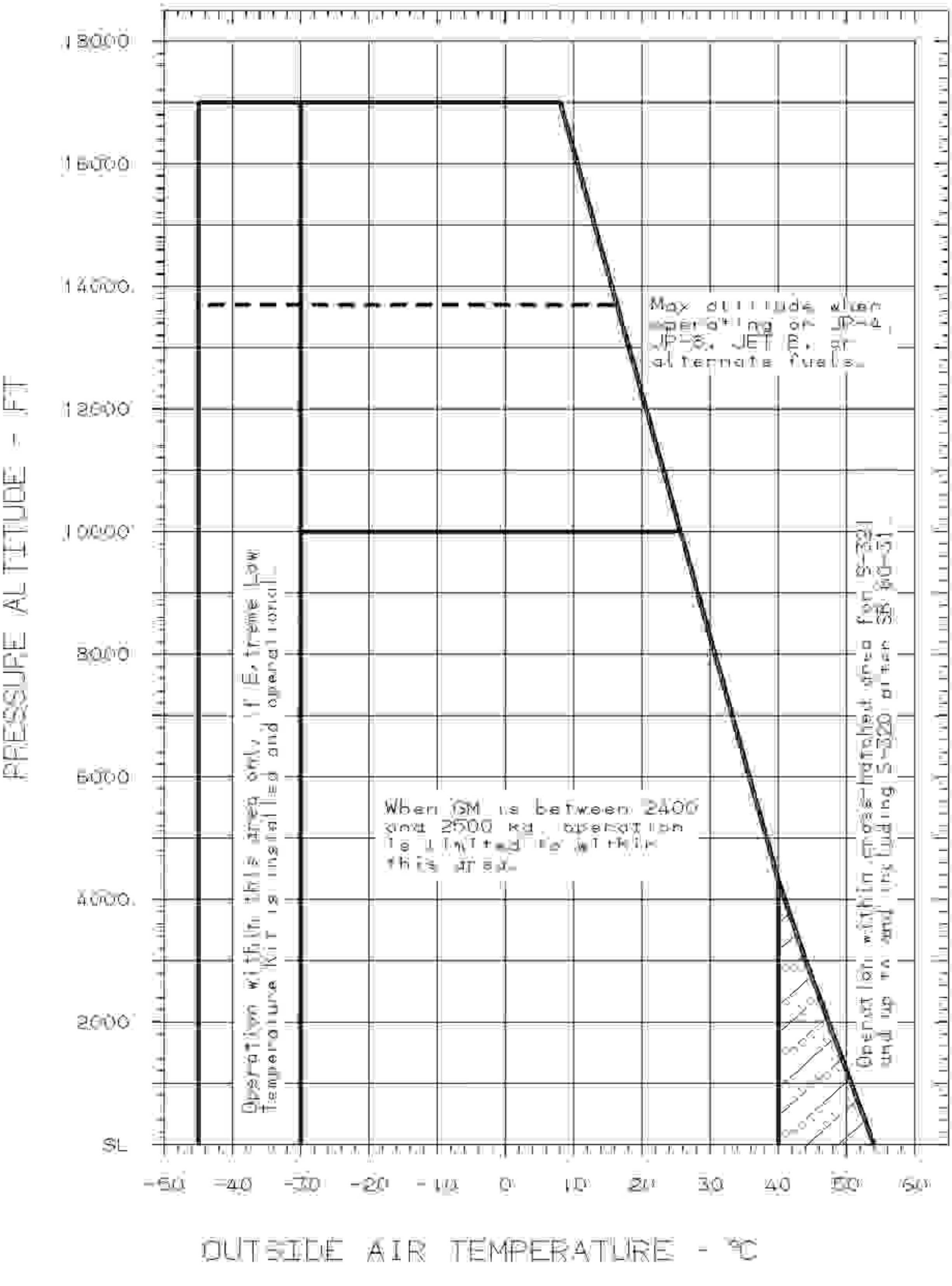


Fig. 2-3 Altitude/Temperature Limit Diagram

2.10 ENVIRONMENTAL OPERATING CONDITIONS

2.10.1 Altitude/Ambient Air Temperature Limitations (see Fig. 2-3)

Min ambient air temperature .... –30 °C

EFFECTIVITY *If extreme low temperature kit 105-80006 installed and operational*

Min ambient air temperature .... –45 °C

EFFECTIVITY *0001-0320 Before SB 60-31*

Max ambient air temperature .... +40 °C

EFFECTIVITY *0321-9999, and 0001-0320 After SB 60-31*

Max ambient air temperature .... +54 °C

When operating in high ambient temperatures (+45 to +54 °C) at sea level, the oil temperatures must be closely monitored in particular during hover. The oil temperatures may be reduced at all times by either landing or transition to cruise flight.

EFFECTIVITY *ALL*

2.10.2 Engine Anti-icing

Engine anti-icing must be on when the ambient temperature is below +4 °C and visible moisture is present, except for takeoff and landing.

2.10.3 Pitot Heating

Pitot heating must be on when the ambient temperature is below +4 °C and visible moisture is present.

2.10.4 Icing Conditions

Flight into known icing conditions is prohibited.

2.10.5 Snow Conditions

Operation in snow is prohibited except when the Continuous Ignition System (see FMS 11-3) is installed and switched on.

The engine inlet deflector shield must be installed when operation in snow is expected.

After operation in snow, a post flight visual check of the compressor inlet must be conducted in accordance with Section 11, FMS 11-3, paragraph 4.4.

NOTE Record engine operation in snow in the journey logbook.

2.10.6 Battery Temperature Limitations

When the engines are started using the onboard battery at an OAT of below –30 °C, the battery shall be preheated to at least –30 °C.

When the engines are started using an external power unit at an OAT of –40 °C or below, the battery shall be preheated to at least –40 °C.

2.11 ROTOR RPM LIMITATIONS

EFFECTIVITY After ASB BO105–60–110 and appropriate placard installed (refer to page 27)

EFFECTIVITY *Variants CB, CB-2 and CDN-B*

	Power ON	Power OFF
Minimum Continuous	98%	85%
Maximum Continuous	102%	110%
Minimum Transient	95%	—
Maximum Transient (15-sec limited)	105%	—

EFFECTIVITY *Variants CBS, CBS-2 and CDN-BS*

	Power ON	Power OFF
Minimum Continuous	98%	85%
Maximum Continuous	102%	104%
Minimum Transient	95%	—
Maximum Transient (15-sec limited)	105%	110%

EFFECTIVITY ALL

EFFECTIVITY ***After ASB BO105–60–110 and appropriate placard not installed (refer to page 27)***

EFFECTIVITY *Variants CB, CB-2 and CDN-B*

	Power ON	Power OFF
Minimum Continuous	95%	85%
Maximum Continuous	102%	110%
Maximum Transient (15-sec limited)	105%	—

EFFECTIVITY *Variants CBS, CBS-2 and CDN-BS*

	Power ON	Power OFF
Minimum Continuous	95%	85%
Maximum Continuous	102%	104%
Maximum Transient (15-sec limited)	105%	110%

EFFECTIVITY *Variants CB, CB-2, CDN-B, CBS, CBS-2 and CDN-BS*

NOTE Maintain rotor RPM between 98-102% whenever operating at airspeeds between V_{NE} minus 20 KIAS and V_{NE}.

EFFECTIVITY *Variants CB-4 and CDN-B-4*

	Power ON	Power OFF
Minimum Transient	95%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	110%
Maximum Transient (15-sec limited)	105%	—

EFFECTIVITY *Variants CBS-4 and CDN-BS-4*

	Power ON	Power OFF
Minimum Transient	95%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	104%
Maximum Transient (15-sec limited)	105%	110%

EFFECTIVITY *ALL*

2.12 ENGINE AND TRANSMISSION POWER LIMITATIONS

Engines: Allison 250-C20 B

Transmission: Zahnradfabrik Friedrichshafen AG (ZF) FS 72B

CAUTION
- OEI EMERGENCY POWER RATING IS LIMITED TO USE ONLY AFTER ACTUAL FAILURE OF AN ENGINE.
- FOR OEI TRAINING OR DEMONSTRATION PURPOSES MAXIMUM CONTINUOUS POWER RATING (OEI) SHALL NOT BE EXCEEDED.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N_1 %	MAX N_2 %
Start/shutdown Transient (Max 10 sec above810°C)		927		
Transient (Max 6 sec above 810°C)		843		
Transient (Max 15 sec)			106	105
All Engines Operating				
Max Continuous Power	2 x 86	779	105	102
Takeoff Power	2 x 86	810 [1)]	105	102
One Engine Inoperative				
Max Continuous Power	95	779	105	102
Emergency Power	95	810 [2)]	105	102

NOTE [1)] Takeoff Power TOT range of 779-810 °C has a 5-min limit.

[2)] Emergency Power TOT range of 779-810 °C is time limited depending on the certifying authority, i.e.

- 5 min for FAA and RAI certified helicopters
- 30 min for LBA and TCAG certified helicopters

2.13 OTHER ENGINE AND TRANSMISSION LIMITATIONS

2.13.1 Engine Oil Pressure

Below 79% N_1 speed .... 3.5 to 9.1 kp/cm^2

From 79 to 94% N_1 speed .... 6.3 to 9.1 kp/cm^2

At 94% N_1 speed and above .... 8.1 to 9.1 kp/cm^2

During engine starting, a positive oil pressure indication must be obtained upon reaching 59% N_1 (Idle) speed.

NOTE During cold weather operation, 10 kp/cm^2 engine oil pressure is allowable following an engine start. When the 9.1 kp/cm^2 limit is exceeded, operate engine at minimum power (IDLE) until normal oil pressure limits are attained.

2.13.2 Engine Oil Temperature

Minimum during start .... –30 °C

Normal operating range .... +30 to +107 °C

NOTE Engines may be operated at oil temperatures from 0 to 107 °C provided engine oil pressure is within specified limits.

EFFECTIVITY *If extreme low temperature kit 105-80006 installed and operational*

Normal operating range .... 0 to +107 °C

EFFECTIVITY *ALL*

Maximum .... +107 °C

2.13.3. Engine Starter

To prevent starter overheat damage, limit starter energize time to the following:

External Power			Onboard Battery		
20 seconds	-	ON	30 seconds	-	ON
60 seconds	-	OFF	60 seconds	-	OFF
20 seconds	-	ON	30 seconds	-	ON
60 seconds	-	OFF	60 seconds	-	OFF
20 seconds	-	ON	30 seconds	-	ON
30 minutes	-	OFF	30 minutes	-	OFF

The 30-minute cooling period is required before beginning another starting cycle.

The starter energize time is the time that elapses between initiation of the starter and ignition in the turbine.

<u>Motoring condition</u>

External Power			Onboard Battery		
15 seconds	-	ON	20 seconds	-	ON
60 seconds	-	OFF	60 seconds	-	OFF
15 seconds	-	ON	20 seconds	-	ON
23 minutes	-	OFF	23 minutes	-	OFF

2.13.4 Generator

Maximum load per generator .... 150 A

2.13.5 Ground Power Starts

The current flow during starting engines shall not exceed 600 A when using 28V DC ground power units.

2.13.6. Main Transmission Oil Pressure

Minimum .... 0.5 kp/cm^2

Normal range .... 0.8 to 2.0 kp/cm^2

NOTE During engine starting, the oil pressure may exceed the normal range.

2.13.7 Main Transmission Oil Temperature

Minimum during start .... –30 °C

Normal operation range .... +30 to +105 °C

EFFECTIVITY *If extreme low temperature kit 105-80006 installed and operational*

Normal operation range .... 0 to +105 °C

EFFECTIVITY *ALL*

Maximum .... +105 °C

Transient (10-min limited) .... +120 °C

2.14 FUEL LIMITATIONS

2.14.1 Fuel Pressure

Normal operating range .... 0.6 to 1.7 kp/cm^2

Maximum .... 1.7 kp/cm^2

2.14.2 Fuel Specifications

Fuel conforming to the following specification is authorized for use:

Type Fuels (Primary)

Wide-cut Type	- MIL-T-5624, grade JP-4	*)
	- ASTM D-1655, Jet B	
	- CAN 2-3.22	
Kerosene Type	- MIL-T-5624, grade JP-5	*)
	- MIL-T-83133, grade JP-8	*)
	- ASTM D-1655, Jet A or Jet A1	
	- CAN 2-3.23	
	- GOST #10227, TS-1 (Regular or Premium)	
	- GOST #16564, RT	

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

Type Fuels (Emergency)

CAUTION MIL-G-5572 FUEL CONTAINING TRICRESYLPHOSPHATE (TCP) ADDITIVE SHALL NOT BE USED.

MIL-G-5572E and subsequent, all grades without TCP

NOTE Maximum of 6 hours operation per turbine overhaul period.

2.14.3. Fuel Specifications - Cold Weather

WARNING AT AMBIENT TEMPERATURES BELOW 4 °C, SOME TYPE OF FUEL ANTI-ICING ADDITIVE IS REQUIRED.

To assure consistent starts at ambient temperatures below 4 °C, the following fuels are recommended.

Type Fuels (Primary)

Wide-cut Type	- MIL-T-5624, grade JP-4	*)
	- ASTM D-1655, Jet B	

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

NOTE
- Jet A, Jet A1, JP-5 or JP-8 may start the engines at temperatures below 4 °C; however, when cold soaked, marginal starts may result due to viscosity changes. Once started, the engines will operate satisfactorily on JP-5, JP-8, Jet A and Jet A1 at fuel temperature and OAT down to –32 °C.
- GOST specification fuels must have a minimum viscosity of 6 centistokes when starting cold engines under low temperature conditions.

Type Fuels (Alternate)

CAUTION
- JP-4 OR JET B FUEL SHALL NOT BE MIXED WITH AVGAS.
- THIS HELICOPTER MAY NOT BE OPERATED WITH AN AVGAS/JET FUEL MIXTURE WHEN THE OAT IS 4 °C OR HIGHER.

AVGAS/Jet A, Jet A1, JP-5 or JP-8 (MIL-T-83133) mixture

The AVGAS/Jet fuel mixture is defined as:

– One part by volume AVGAS (MIL-G-5572) of either grade 80/87 or grade 100/130 (100L) having 0.53 ml/l maximum lead content. Do not use grade 100/130 having 4.6 ml/l lead content.

– Two parts by volume ASTM D-1655, Jet A or Jet A1, or MIL-T-5624, grade JP-5.

NOTE Use of 100/130 (100L) grade AVGAS/Jet fuel mixture shall be restricted to 300 hours in one engine overhaul period due to the high lead content of the fuel.

2.14.4 Fuel Additives

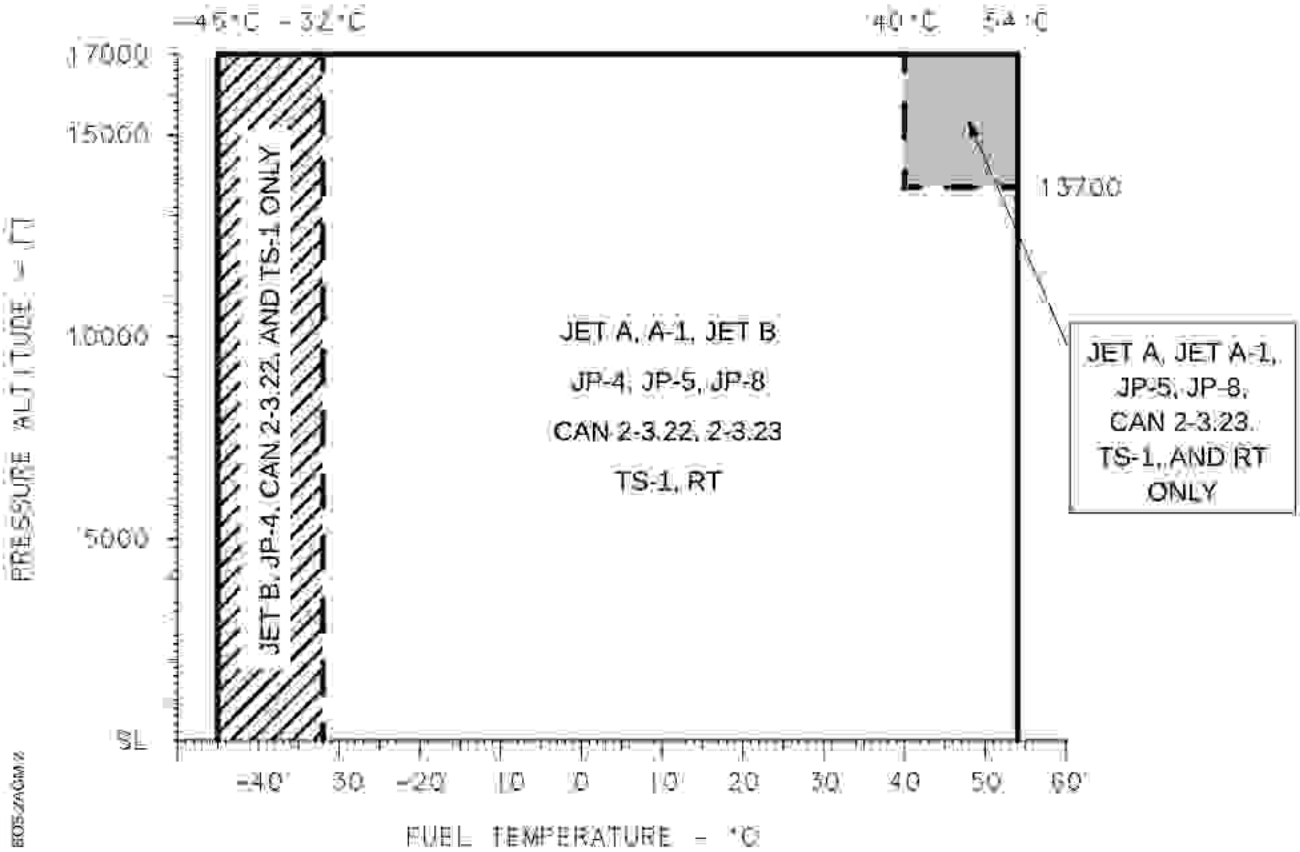
2.14.4.1 Anti-icing

Icing protection is required when operating with primary fuels at temperatures below +4 °C (40 °F).

– MIL-I-27686; Concentration by volume: Max 0.15 %, min 0.035 %.
– Fluid I (GOST 8313), Concentration by volume. Max 0.3 %, min 0.10 %. *)
– Fluid IM (TU 6-10-1458); Concentration by volume: Max 0.3 %, min 0.10 %. *)

*) Only for use with GOST specification fuels.

2.14.5 Fuel Temperature



NOTE REFER TO HELICOPTER OAT LIMITS

Fig. 2-4 Temperature Limits for Approved Primary Fuels

2.14.6 Fuel Quantities

TANK	USABLE FUEL		UNUSABLE FUEL	
	liters	kilograms	liters	kilograms
Main	477	382	7.5	6.0
Supply	93	74	2.5	2.0
Totals	570	456	10.0	8.0

Fuel mass values are based on a fuel density of 0.8 kg/l.

The usable fuel quantity is empty when zero is indicated on the relevant fuel quantity gauge.

2.15 OIL LIMITATIONS

2.15.1 Oil Specifications

	Oil Type	OAT Limit
Engines	MIL-PRF-23699 MIL-PRF-7808	–30 °C –30 °C
Main Transmission	MIL-PRF-23699	-30 °C
Intermediate Gearbox	MIL-PRF-23699	–45 °C
Tail Rotor Gearbox	MIL-PRF-23699	–45 °C
Main Rotor Hub	MIL-PRF-23699	-45 °C

2.15.2 Oil Quantities

	Liters	Kilograms
Engines (each tank)	4.5	4.5
Main Transmission	10.3	10.3
Intermediate Gearbox	0.6	0.6
Tail Rotor Gearbox	0.4	0.4
Main Rotor Hub	1.8	1.8
Oil mass values are based on an oil density of 1 kg/l.		

2.16 HYDRAULIC SYSTEM LIMITATIONS

2.16.1 Operational Hydraulic System

For takeoff the operational hydraulic system must be system 1. A change-over to system 2, by operating the hydraulic test switch during flight, is prohibited. The hydraulic test switch is for ground and preflight test purposes only.

2.16.2 Hydraulic Fluid

Hydraulic fluid type MIL-H-5606 is authorized for use at all ambient temperatures.

Hydraulic fluid quantity per system .... 1.2 liters

2.17 OPERATIONAL LIMITATIONS

2.17.1 Rotor Starting and Stopping in Wind

Rotor starting and stopping has been demonstrated with up to 45 KTAS headwinds.

2.17.2 Hover Turn

Maximum rate .... 90° per sec

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 <u>when GM is between 2400 and 2500 kg</u>*

Maximum rate .... 45° per sec

EFFECTIVITY *ALL*

2.17.3 Main Rotor Mast Moment

The mast moment indicator shows the bending moment of the main rotor mast. The red radial marking shall not be exceeded during slope landing, flight maneuvers and ground operation.

Maximum .... red radial

Caution range .... yellow arc

Normal range .... green arc

Calibration point .... white radial

NOTE
- A logbook entry and maintenance action are required whenever:
  - the **LIMIT** warning light <u>remains</u> on (even after pointer returns to a lower mast moment), and
  - the helicopter is operated with a defective mast moment system
- Periodical maintenance actions are required according to Maintenance Manual, if the helicopter is operated without the mast moment system.

2.17.4 Slope Operations

Slope operations (takeoff/landing) are limited in the degree of sloping terrain upon which the maneuver may be performed:

– **Mast moment indicator installed and operative**

Ground sloping down to the left .... Max 15°
Ground sloping down to the right .... Max 8°
Ground sloping down forward .... Max 8°
Ground sloping down rearward .... Max 10°

– **Mast moment indicator inoperative or removed**

Ground sloping in all directions .... Max 5°

2.17.5 Prohibited Flight Maneuvers

Aerobatic maneuvers are prohibited.

2.17.6 Mass-altitude-temperature

For mass-altitude-temperature limitations at takeoff, landing, and during ground effect maneuvers refer to Section 5.

2.17.7 Extreme Low Temperature Operation

Operation at extreme low ambient temperatures (below –30 °C) is permitted only under the following conditions:

- Extreme low temperature kit must be installed and operational
- Cockpit/cabin temperatures must be above –30 °C
- Engine and transmission oil temperatures must be above 0 °C prior to takeoff.

Refer to BO105 Maintenance Manual, Chapter 101, "Measures to be taken when operating at extremely low temperatures".

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4 when GM is between 2400 and 2500 kg*

Operation at ambient temperatures below –30 °C is not permitted.

EFFECTIVITY *ALL*

2.17.8 Engine Inlet Deflector Shield

EFFECTIVITY *0001-0450 Before SB 60-37*

The engine inlet deflector shield must be removed when the OAT is above +30 °C.

EFFECTIVITY *0451-9999, and 0001-0450 After SB 60-37*

The engine inlet deflector shield must be installed at all times.

EFFECTIVITY *ALL*

2.18 INSTRUMENT MARKINGS

The pointers and scales of the instruments are marked as follows:

Left engine	1
Right engine	2
Main rotor/transmission	R
Main fuel tank	M
Supply fuel tank	S

Minimum and maximum limits	red radial
Calibration point	white radial
Start and transient limits	red dot
Normal/continuous range	green arc
Start and caution range	yellow arc
Never exceed speed – power-off	blue radial

2.18.1 Airspeed Indicator

10 to 45 kt	yellow arc
45 to 145 kt	green arc
145 kt	red radial
100 kt	blue radial

2.18.2 Engine RPM (N_1) Indicator

59 to 105%	green arc
105%	red radial

2.18.3 **Triple Tachometer**

EFFECTIVITY *Variants CB, CB-2, CDN-B, CBS, CBS-2 and CDN-BS*

Engine RPM (N_2) (power-on limits)

95%	red radial
95 to 102%	green arc
102%	red radial
105%	red dot

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*

Engine RPM (N_2) (power-on limits)

95%	red dot
98%	red radial
98 to 102%	green arc
102%	red radial
105%	red dot

EFFECTIVITY *Variants CB, CB-2, CB-4, CDN-B and CDN-B-4*

Rotor RPM (power-off limits)

85%	red radial
85 to 110%	green arc
110%	red radial

EFFECTIVITY *Variants CBS, CBS-2, CBS-4, CDN-BS and CDN-BS-4*

Rotor RPM (power-off limits)

85%	red radial
85 to 104%	green arc
104%	red radial
110%	red dot

EFFECTIVITY *ALL*

2.18.4 **Turbine Outlet Temperature (TOT) Indicator**

0 to 779 °C	green arc
779 to 810 °C	yellow arc
810 °C	red radial
843 °C	red dot
927 °C	red dot

2.18.5 **Triple Oil Pressure Indicator**

Engine oil pressure

3.5 kp/cm^2 .... red radial
3.5 to 6.3 kp/cm^2 .... yellow arc
6.3 to 9.1 kp/cm^2 .... green arc
9.1 kp/cm^2 .... red radial

Transmission oil pressure

0.5 kp/cm^2 .... red radial
0.5 to 0.8 kp/cm^2 .... yellow arc
0.8 to 2.0 kp/cm^2 .... green arc

2.18.6 **Triple Oil Temperature Indicator**

Engine oil temperature

30 to 107 °C .... green arc
107 °C .... red radial

Transmission oil temperature

30 to 105 °C .... green arc
105 °C .... red radial

EFFECTIVITY *If extreme low temperature kit 105-80006 installed*

Engine oil temperature

–30 to 0 °C .... yellow arc
0 to 107 °C .... green arc
107 °C .... red radial

Transmission oil temperature

–30 to 0 °C .... yellow arc
0 to 105 °C .... green arc
105 °C .... red radial

EFFECTIVITY *ALL*

2.18.7 **Dual Torque Indicator**

0 to 86%	green arc
86%	red radial
95%	red radial

2.18.8 **Fuel Pressure Indicator**

0 to 0.6 kp/cm^2	yellow arc
0.6 to 1.7 kp/cm^2	green arc
1.7 kp/cm^2	red radial

2.19 PLACARDS AND DECALS

The placards and decals depicted herein are typical and presented in either the English language or bilingual (German-English). Other languages may be used as a customer option. Slight formal differences from the installed placards and decals do not affect the information presented therein.

EFFECTIVITY *For IAC-AR registered helicopters only*

During flights in metric airspace, check the altitude using the conversion table placard (ft → m and hPa → in. Hg) located on the windshield center post in pilot's field-of-view.

EFFECTIVITY *All*

Text:

THIS HELICOPTER MUST BE OPERATED IN COMPLIANCE WITH THE OPERATING LIMITATIONS SPECIFIED IN THE LBA APPROVED ROTORCRAFT FLIGHT MANUAL	**THE "AIRWORTHINESS LIMITATIONS" SECTION OF THE ROTORCRAFT MAINTENANCE MANUAL MUST BE COMPLIED WITH.**

Location: In pilot's field-of-view

Text:

PRESSURE ALTITUDE-FT	OAT °C									
	-45	-30	-20	-10	0	+10	+20	+30	+40	+50
0	145	145	145	145	145	145	144	141	139	137
2000	145	145	145	145	144	141	139	136	134	132/129
4000	145	145	143	141	139	136	134/132	132/128	129/123	
6000	145	141	138	136	134/132	131/127	129/122	127/118		
8000	140	136	133/132	131/127	129/122	126/117	123/113	119/108		
10000	135/134	131/127	128/122	126/117	122/112	118/107	114/103			
12000	130/124	126/117	123/112	117/107	118/102	108/97	104/93			
14000	124/113	117/106	112/102	108/97	103/92	99/88	NOTE MAINTAIN IN SPEED RANGE FROM VNE MINUS 20KTS TO VNE 98-102% ROTOR RPM			
16000	114/103	107/96	103/91	98/87	93/82	89/78				
17000	109/98	102/91	98/86	93/81	88/77					
	VNE-KIAS GROSS WEIGHT (kg) 2100/2400									

BO5-2BCM_0

Location: In pilot's field-of-view

EFFECTIVITY *Variants CB-4, CDN-B-4, CBS-4, and CDN-BS-4*

Text:

For gross weights between 2400kg and 2500kg: V_{NE}=$V_{NE\ 2400kg}$−10%
Rotor RPM. 98% to 102% at any speed

Location: In pilot's field-of-view

EFFECTIVITY *ALL*

Text:

DURING GROUND OPERATION ONLY SMALL CYCLIC STICK DISPLACE-MENTS FOR FUNCTIONAL TESTS

Location: In pilot's field-of-view

Text:

MAX. BODENLAST 600 kg/m^2 LADEGUT VERZURREN ZUL. BELASTUNG PRO VERZURRÖSE 100 kg ZUL. BELASTUNG PRO GEWINDEEINSATZ 200 kg MAX. WEIGHT PER sq. ft. 120 lbs CARGO TO BE SECURED MAX LOAD PER EYE 220 lbs MAX LOAD PER INSERT 440 lbs

Location: Cargo compartment

EFFECTIVITY *RAI certified helicopters only*

Text:

MAX. CARGO IS LIMITED BY WEIGHT AND BALANCE CONSIDERATION

Location: Cargo compartment

EFFECTIVITY *ALL*

Text:

MAX ZULADUNG	20kg
MAX LOAD	44lbs

Location: Avionics equipment bay

Text:

BEI FASS-ODER KANISTERBETANKUNG EINGEBAUTES SIEB VERWENDEN
FOR BARREL OR GAS CAN REFUELING USE INSTALLED SCREEN
BEI BETRIEB UNTER +4°C ENTEISUNGSZUSATZ NACH MIL-I-27686
VOLUMENANTEIL: MAX. 0,15%; MIN. 0,035%
FOR OPERATION BELOW +4°C (40°F) ADD ANTI-ICING ADDITIVE MIL-I-27686
CONCENTRATION BY VOLUME: MAX. 0,15%; MIN. 0,035%

Location: Adjacent fuel tank filler neck

Text:

KRAFTSTOFF	570 LITER
FUEL CAP.	150 US GAL
MIL-T-5624:	JP-4, JP-5
MIL-T-83133:	JP-8
ASTM-D-1655:	JET A, A-1, B

Location: Adjacent fuel tank filler neck

Text:

4l — max — 4l
Oil-System
4,5l (1,2 US Gal)
nicht
mischen!
do not mix!
3l — — 3l
Nachfüllen
Add Oil
2l — MIL-L-23699 — 2l
(MIL-L-7808F,G)

Location: Engine oil tank

Text:

ÖL 10,3 LITER
OIL 2.7 US GAL
MIL-L-23699

Location: Adjacent main transmission filler neck

Text:

TÜR-NOTABWURF
TÜRE ÖFFNEN
DANN ABWURFHEBEL
BETÄTIGEN
DOOR JETTISON
OPEN DOOR
THEN PUSH DOWN
EMERGENCY LEVER

Location: Interior each cockpit door – above door handle

Text:

SCHIEBETÜR
ÖFFNEN OPEN
DRÜCKEN PUSH
SCHIEBEN SLIDE
SLIDING DOOR

Location: Interior LH sliding door – above door handle

Text:

SCHIEBETÜR
ÖFFNEN OPEN
DRÜCKEN PUSH
SCHIEBEN SLIDE
SLIDING DOOR

Location: Interior RH sliding door – above door handle

EFFECTIVITY After ASB BO105–60–110 and appropriate placard installed

Text:

MIN CONTINUOUS 98%
MIN TRANSIENT 95%
ASB BO105 60–110

Location: Next to the triple RPM indicator

EFFECTIVITY ALL